



Fig. 2: Hot aisle/cold aisle with hot air above dropped ceiling

temperatures mean a lower dew point or a point where humidity in the air will condense. This can cause problems in a data centre that has an uneven heat load, causing the various CRAC units in the data centre to operate at different temperatures. Often, one set of CRAC units will expend energy providing humidity to the room, while another set, located in a different area, works at removing humidity from the air as it tries to maintain a lower temperature to compensate for a high-density section of the room.

The solution to cooling high-density data centres varies marginally between the various cooling manufacturers, and the general approach is the same: Efficiently get the hot air coming out of the back of the racks to a cooling system without allowing it to mix with the cool air on the front side of the racks. All the solutions use a variation of hot-air/hot-aisle containment.

The four cooling options are:

**Option 1.** Create a hot-air return space above a raised ceiling. Each of the racks is run through ducts to the return space. The intake ducts on the CRAC units are extended up to the raised ceiling level so that these receive all of their intake air directly from the hot aisle without mixing of the hot and cold air. This results in producing hot air above the raised ceiling and then

routing it directly to CRAC units.

**Option 2.** Create a microenvironment made up of two rows of racks. Each pair of racks contains at least two in-row CRAC units capable of cooling the hot air being exhausted from the back of the racks. Typically, the hot aisle is sealed from the rest of the computer room, once again ensuring that the hot air and cold air do not mix up.

**Option 3.** It's a less strict version of Option 2 in that small CRAC units are suspended from the ceiling or mounted on top of the racks that use the hot-aisle air for their intake and supply cold air back to the cold aisle. The hot aisle is not sealed as in the first two options. So it is necessary to ensure that the overhead units are capable of enough airflow to ensure negative pressure in the hot aisle so that the hot air does not drift away around the ends of the racks or over the top of the racks.

**Option 4.** Reduce the microenvironment to the rack level. This allows for high-density cooling at the rack level with redundancy.

All these options allow for adequate high-density cooling. The selection process for identifying the most cost-effective solution for any particular data centre environment depends on the site-specific details. An additional benefit of the hot-air containment solution is that the rest of

the data centre can be maintained at a comfortable working temperature. Because the hot air is captured and contained, the data centre maintains an even temperature throughout, eliminating the scenarios where various CRAC units work against each other or are constantly cycling 'on' and 'off.'

**Power cabling.** Providing power to the various racks of x86 servers in the past was fairly simple. Install a pair of redundant 20A, 110V circuits, get a couple of rack-mount power strips and plug in your servers. You would run out of space in the rack before running out of power.

With higher-density servers and change in rack sizes, a flexible power distribution system for data centres has become more than a nice-to-have option. Products are available that provide a flexible, reconfigurable system for power distribution without having to run conduit in most areas of the country.

Other than the capacity of the power supply and battery backup system, one of the biggest limiting factors on data centre flexibility is the sheer quantity of circuit breakers required. For example, a rack that used to take two 20A, 110V circuits would require one single-pole circuit breaker in each circuit breaker panel of a redundant system. On a panel that supports 42 circuit breakers, you could support 42